



Q-MAP

Team 5

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Team Charter

Purpose & Key Responsibilities

Purpose: Our purpose is to help students, faculty, the public, etc. navigate across all 3 campuses for classrooms, bathrooms, or any other essential location that needs to be found. Our mission is to utilize the current campus maps and resources to find the most efficient way to get to a specific location. Connect your QU ID and it will give you the best route to finding where you need to go.

Key Responsibilities: As a team, it is our responsibility to ensure that our team assignments are completed in a timely manner with our best efforts, ensuring a quality product that will allow users to easily find their way around campuses to their desired location. All team members are expected to work proactively and productively. Feedback will be welcomed.

Vision

To make an easy-to-use app that provides a personalized navigation and information experience for the Mount Carmel and York Hill campuses. We aim to provide students, faculty, and others with a helpful tool to use when navigating Quinnipiac campuses.

Values

Our team values the equality of time efficiency when it comes to completing the tasks on time and to our best efforts to make this project as easy and flowing as possible. We also value each of our own individual ideas and respect that we are all here to collaborate and produce one unison idea for a successful app. We uphold honesty and integrity for all our asks in getting to an outcome.

Goals

Our overall goals are to complete the assignment and each task leading up to the final submission on time and complete. Creating an app that is helpful for all that are on a Quinnipiac campus, where it makes this problem less of an issue

Roles & Responsibilities

Everyone will be responsible for completing specific tasks and goals by a certain time. Additionally, we aim to ensure that all team members contribute equally.

Mutual Expectations/Performance Evaluations

Ensure that all team members are working efficiently and that all tasks and goals are completed by a specific and set timeframe.

Operating Procedures

Team Meetings/Communication: The team will meet 1-2 times a week, fluctuating depending on what work needs to be done. If work is not completed during the class period, the team will discuss via iMessage group

chat where to meet at what time. If a team member cannot meet during the designated time, they are responsible for informing the other members via text as soon as possible. Members are expected to respond to any communication within 24 hours.

Decision Making: Work will be distributed evenly and will be agreed upon by the team at the start of work. If any conflict arises, the work can be redistributed, but all members are expected to contribute an equal amount. As a team, members are expected to review the entirety of the submission before submitting.

Conflict: If there becomes a conflict between two or more group members, then we will try to figure out what is best for our app overall. If both sides are extremely passionate about their own opinions and there is no resolution, then we will try to incorporate both ideas into our project. If this still does not satisfy the group, then we will involve the professor and see what the best possibility is moving forward for our group.

Reflection: Establishing clear objectives for each member of our group to achieve during the specified time period. Ongoing communication to discuss performance, concerns, and feedback.

Executive Summary

The proposed Q-Map application is designed as a resourceful, real-time navigation tool for Quinnipiac's Mount Carmel Campus, operated by students to provide accurate, up-to-the-minute details on classroom locations, amenity status, and optimized routes. This project presents a strong opportunity to integrate with student course schedules and gain user trust through official Quinnipiac branding, with initial financing secured by the university and operational costs potentially offset by in-app advertisements. Technically feasible using cellular radar and GPS, the app must overcome threats like the preference for preexisting apps (e.g., Google Maps), user concerns over data sharing, and potential loss of trust if real-time updates are slow or if certain campus areas have poor network coverage. The overall development schedule is ambitious, requiring an estimated 14 month, including four months dedicated solely to comprehensive, room-level data collection, with the initial design and advanced feature development costs estimated between \$15,000 and \$270,000, plus ongoing annual maintenance and support.

Introduction

Navigating the dynamic Mount Carmel Campus is currently inefficient, leading to wasted time and frustration when searching for indoor locations. General navigation applications simply fail in this critical environment. The solution to students' frustration is Q-Map, a cutting-edge, student-developed indoor navigation application uniquely tailored for Quinnipiac. Leveraging the intimate, real-time knowledge of our student team, Q-Map guarantees superior accuracy and immediate relevance by providing precise, GPS-backed locations for every classroom, office, and amenity, along with essential real-time updates on bathroom maintenance status and water fountain availability.

The app's core benefit is its seamless integration with the student experience. Q-Map will sync directly with each student's course schedule or planner, automatically calculating and displaying the most efficient and convenient route to their next destination. Technically, this is achieved by utilizing a combination of GPS and a cellular radar system to determine a user's exact location, therefore providing the most effective route based on comprehensive, previously collected data. By delivering this level of highly localized detail, Q-Map offers a user experience vastly superior to generic alternatives, ensuring every student gets where they need to be faster and with confidence.

Q-Map is designed to be both feasible and scalable, representing a sound investment in campus technology leadership. While initial development is funded by Quinnipiac University, our long-term economic sustainability is secured through the integration of in-app advertisements, providing a clear path to cover operational costs and support future expansion to the York Hill and North Haven campuses. The app drastically improves student life, and we maintain robust user privacy, requiring acceptance of terms and conditions that ensure location tracking only occurs when the app is actively open. Investment in Q-Map is a strategic move toward enhancing student efficiency and delivering a superior Quinnipiac experience.

Business Case

SWOT Analysis

Strengths:

The Q-Map application boasts significant strengths by being run by current Quinnipiac students who possess intimate knowledge of the Mount Carmel Campus layout, ensuring highly relevant and accurate guidance. Its primary value lies in providing real-time updates on precise locations for classrooms, bathrooms, and water fountains, as well as bathroom maintenance status, and syncing with campus maps to deliver the most convenient route to a user's planned destination. This localized, detailed information offers superior user experience compared to general navigation tools.

Weaknesses:

Key weaknesses primarily revolve around adoption and maintenance challenges. Students who choose not to use the app, particularly during the beginning of a new semester, will still face difficulty navigating classroom changes. Furthermore, the app's success is highly dependent on the speed and reliability of its data updates; if the information isn't refreshed quickly enough, users will lose trust in the app's accuracy. Finally, the application's functionality could be hindered by poor internet or GPS coverage in specific areas of the large campus.

Opportunities:

There are several compelling opportunities for Q-Map to thrive. The app can significantly enhance student life by working alongside each student's course planner/schedule to automatically identify classroom locations. By offering an easy-to-use, real-time navigation app uniquely tailored to the campus, it differentiates itself from existing solutions. Official Quinnipiac branding and a commitment to live updates will support user trust and reliability. Additionally, the integration of in-app advertisements provides a viable model to cover potential operational costs and contribute to sustainability.

Threats:

The project faces significant threats from established competitors, primarily preexisting navigation apps like Google Maps/Google Earth, which users may prefer due to familiarity. Addressing user concerns over tracking and shared data usage is critical, as the app requires location sharing to function. Moreover, the accuracy and timeliness of reporting campus closures and maintenance could be limited, potentially frustrating users and undermining confidence in the app's utility.

Cost-Benefit Analysis:

The development of the Q-Map application presents a varying range of potential costs depending on the desired feature set. A basic app with core functionalities like real-time location tracking and basic navigation is estimated to cost between \$10,000 and \$50,000, while an advanced app incorporating sophisticated features like route optimization and geofencing could cost significantly more, from \$50,000 to \$250,000. Additional key

investments include UI/UX design (\$5,000 to \$20,000) and the integration of AI/ML for predictive routing (\$25,000 to \$50,000). Beyond development, there are annual costs for maintenance and support (\$10,000 to \$30,000 per year) and monthly expenses for marketing and promotion (\$1,500 to \$32,000). While the server and infrastructure costs appear prohibitively high, the benefit of providing students with an easy-to-use, reliable, and real-time navigation tool specifically tailored to the Mount Carmel campus significantly enhances operational efficiency, reduces new-student anxiety, and improves the overall Quinnipiac campus experience.

Feasibility Analysis:

Technical:

This application will be a resourceful tool that helps students, faculty, and guests of Quinnipiac's Mount Carmel campus have a better understanding of where any destination on campus is and how to get to it from that specific user's location. This will be in the form of GPS and using a cellular radar system to find out exactly where the user is. Then, from previously collected data on each room/bathroom/food/office location, the app will provide the most efficient and effective way to get to that precise location.

Economics:

The development of the app will be funded by Quinnipiac University. As the app profits and makes money back on the number of users and Quinnipiac paying for each user to use this app, then we may expand our horizons in York Hill and the North Haven Campus.

Legal/Contractual:

Every student must accept the terms and conditions of the application. This will allow for the locations of the phones to be shared to the app and get the exact coordinates of where the user is to find the best route to a destination. To use the app, users must allow for location sharing only when the app is open; otherwise, the app is nonfunctional.

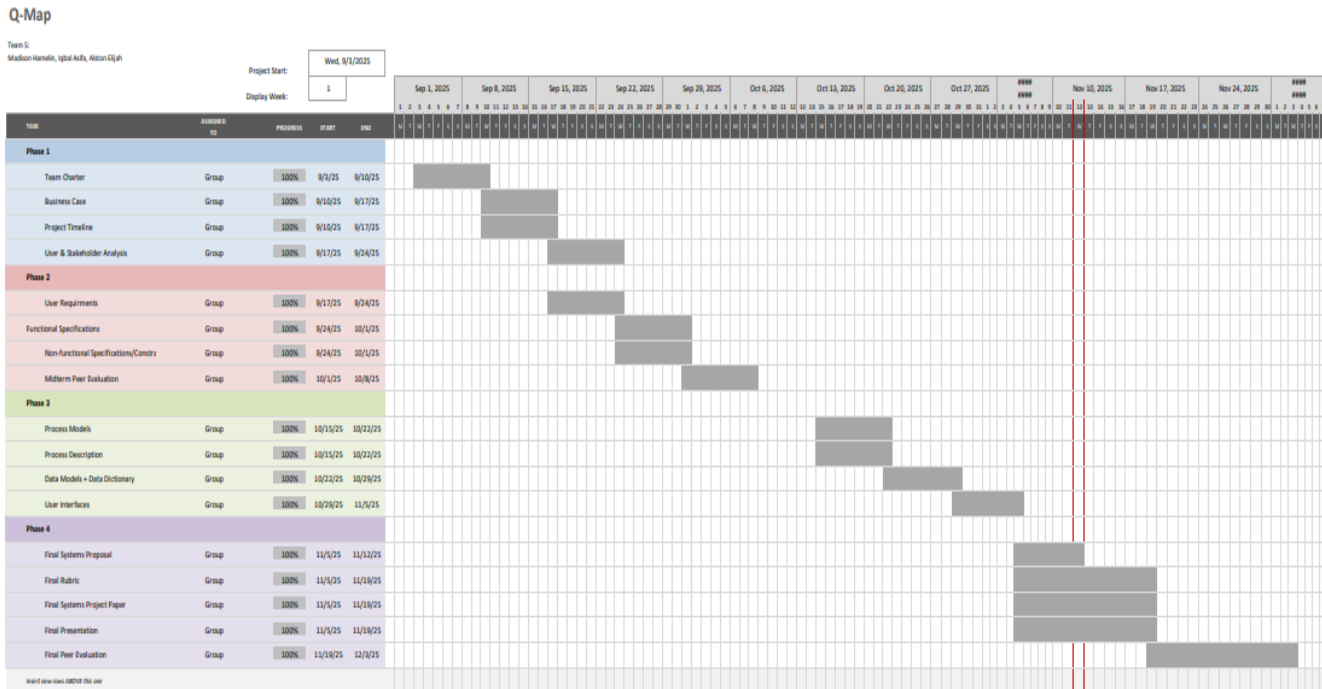
Operational:

This application will make students' lives much easier as they try to find classes in the first few weeks of school, or any user who is looking for a precise location that they are unaware of. It provides a GPS-like system to users on the Mount Carmel

Schedule:

This software will take a while to find all data in every single room and space that is on Mount Carmel, since there is no previous data on this topic for a GPS system. We will need to make sure that all buildings, floors, and rooms are accounted for, especially in the newer buildings that have vastly changed the layout of Mount Carmel campus. This data collection will take approximately 4 months to complete check. The software then must be designed, which could take around 5 months to complete. The implementation process and testing before full usage will take another 5 months. Overall, the start-to-finish will end up taking around 14 months to complete and have the app fully up and running.

Project Timeline



The Gantt chart for the “Q-Map” project outlines the entire project timeline across four main phases: Planning (in blue), Requirements (in red), System Design (in green), and Final Deliverables (in purple). Each phase includes specific tasks assigned to the group, with progress indicated as a percentage. The horizontal calendar bars visually represent the start and end dates for each task. According to this Gantt chart, this chart was accessed on November 12th, and all phases are completed. This chart also shows the overlapping activities and dependencies, making it easier to understand and take apart each phase.

This chart is essential as a project management timeline tool, allowing the team to monitor progress, stay aware of deadlines, and assign accountability. It provides a clear overview of what is completed, what is in progress, and what is coming next. By having a visual timeline, the team can quickly identify if they are on track or falling behind. Overall, the Gantt chart supports efficient planning, communication, and execution throughout the lifecycle of the project.

User & Stakeholder Analysis

Purpose of the System:

We intend to allow users to navigate the Mt. Carmel Campus on our application from their location to their designated location. Information on what areas that are open or closed, may that be pertaining to bathrooms, food options, study rooms, etc. will be provided. GPS navigation and directions on timing and when they should depart to arrive at their specific destination will be shown. An overall map of what is seen on campus and a search bar to select what is needed by the user will be provided.

Primary Users:

Firstly, students can use our application to navigate all buildings, floors, and rooms on campus that are available to them with real-time location tracking. Faculty can also use our application to aid students and visitors by providing real-time information about rooms, class schedules, office hours, and any possible cancellations. Visitors may also use our app for easy navigation to key campus locations and access real-time event information such as schedules, venue details, and any possible changes.

Other Stakeholders:

The Quinnipiac IT Department will be able to help with all aspects of connections to devices and ensure that we get the best possible coordinates of each user's device. They will also be able to make sure that there is safety and security information about users' locations while using the application. The application will help technology be seen as more effective and efficient compared to other Quinnipiac services that can currently be found online. Additionally, the application will require there to be funding from Quinnipiac University to support collection of locations and to make sure that our application can be used on campus. We require these figures to ensure approval for the launch of our application.

User Requirements

Requirements for Students:

We will allow students to navigate from their current location to any building or facility on Mt. Carmel Campus with step-by-step directions. We intend to provide students with real-time updates on whether bathrooms, dining options, and study rooms are open or closed. We also want to give students GPS-like directions that include travel time and reminders for when to leave to arrive at their destination on time. We will provide an overall interactive campus map with a search bar so students can quickly find specific locations. Lastly, we want to allow students to use location services for accurate tracking and navigation on campus.

Requirements for Faculty:

Our app will sync with the faculty's teaching or meeting schedule, providing reminders for their next class or meeting location. It will allow faculty to receive precise navigation from the faculty member's current location to their assigned classroom, office, or campus department. It will alert faculty members of any changes or cancellations that may occur due to ongoing maintenance or closures that may affect the availability of certain locations.

Requirements for Visitors:

Our app will provide directions and information for key locations on campus, such as visitor parking, main office locations, or campus events. It will alert visitors to restricted areas on campus and provide guidance on which locations are open to the public. It provides visitors with detailed information including hours of operation, services offered, and contact information for specific offices and other locations. Additionally, it will offer a map highlighting paths, ramps, elevators, and restrooms that are accessible to all visitors.

Requirements for QU IT Department:

There will be GPS or indoor positioning services to track user location on Mt. Carmel Campus. It will support turn-by-turn routing between buildings and designated destinations. The system should connect campus databases to retrieve real-time facility availability (e.g., bathrooms, dining hours, study room reservations) and ensure accuracy. It will calculate estimated travel times based on distance and average walking speed. It will include a search function that can query facilities, services, and building names. The system should protect user location data with encryption, and the app shall comply with university IT and data privacy policies.

Requirements for QU President and CFO:

We will maintain the app within the approved development and maintenance budget. It will allow students to use QU logins to set up with class schedules. It will supply usage analytics and regular performance reports for financial and operational decision-making. We shall align the app's functionality with the university's mission to support student success and engagement

Functional Requirements

Display Login Screen:

Users will select either student, faculty, or visitor login. After their selection, the space to enter login credentials will be displayed.

Display Visitor Screen:

Display guides for campus buildings, offices, and other locations will be provided. A sync calendar feature will also be provided along with the campus map.

Display Faculty Screen:

Our app will provide faculty with the quickest routes to buildings, rooms, and offices they intend to travel to. Additionally, a sync calendar feature will be available to use.

Display Alert Screen:

Real-time updates on closures of bathrooms, dining areas, and study rooms will be readily available to all users. Our app will display timely travel time alerts prompting users to leave in order to arrive at their destination on time. Relevant operating hours for buildings, offices, or services tied to the selected location will be shown. Emergency alerts such as weather warnings, hazardous conditions, or reported animal sightings will be promptly provided to users.

Display Map Screen:

The app will display the users' current location. If a selection is made, a preview of planned routes to classes will be displayed on the map. When the user begins their route, a GPS-like format of the area around the users' current location will be shown with labels on notable buildings.

Display Search Bar Screen:

A search bar to look up specific locations for meeting rooms, bathrooms, classrooms, etc. will be displayed. Underneath the search bar, recent and popular searches among other users will be shown for convenience.

Display Personalized Information Screen:

App settings and preferences such as app colorization, light and dark modes, etc. will be shown to provide students with their own tailored and unique app experience. Account information such as their name, phone number, email address, and school will also be displayed.

Display Help Screen:

A bar with optional questions will be shown. Steps for solving any issues the user has will be displayed. Lastly, a button to connect directly with an IT Department member will be provided to users.

Nonfunctional Requirements

Operational:

Our app and its system should maintain a user-friendly interface with intuitive navigation for students, faculty, and visitors of Quinnipiac University. The system shall integrate with QU's existing alert system for real-time updates and notifications about emergency alerts, class cancellations, or buildings and rooms closures. The system should remain reliably operational throughout the academic year, with minimal service interruptions outside of pre-scheduled maintenance windows. The system should allow for scheduled maintenance without requiring any downtime for the entire service itself. The system must be able to continue functioning at a limited capacity in the event of one or more data sources becoming temporarily unavailable.

Performance:

The login screen should load within 2 seconds of the application being opened. The map screen should immediately display the users' current location and planned route within 3 seconds of the prompt. Search results shall be returned within 2 seconds after the user enters their search query. The application should be available at all times during the academic school year, as well as updates for new locations. The travel time and GPS system should remain accurate compared to real walking times collected through previous data and usage. Our system shall support 5,000 using the application at once without having any negative effect on response time.

Security:

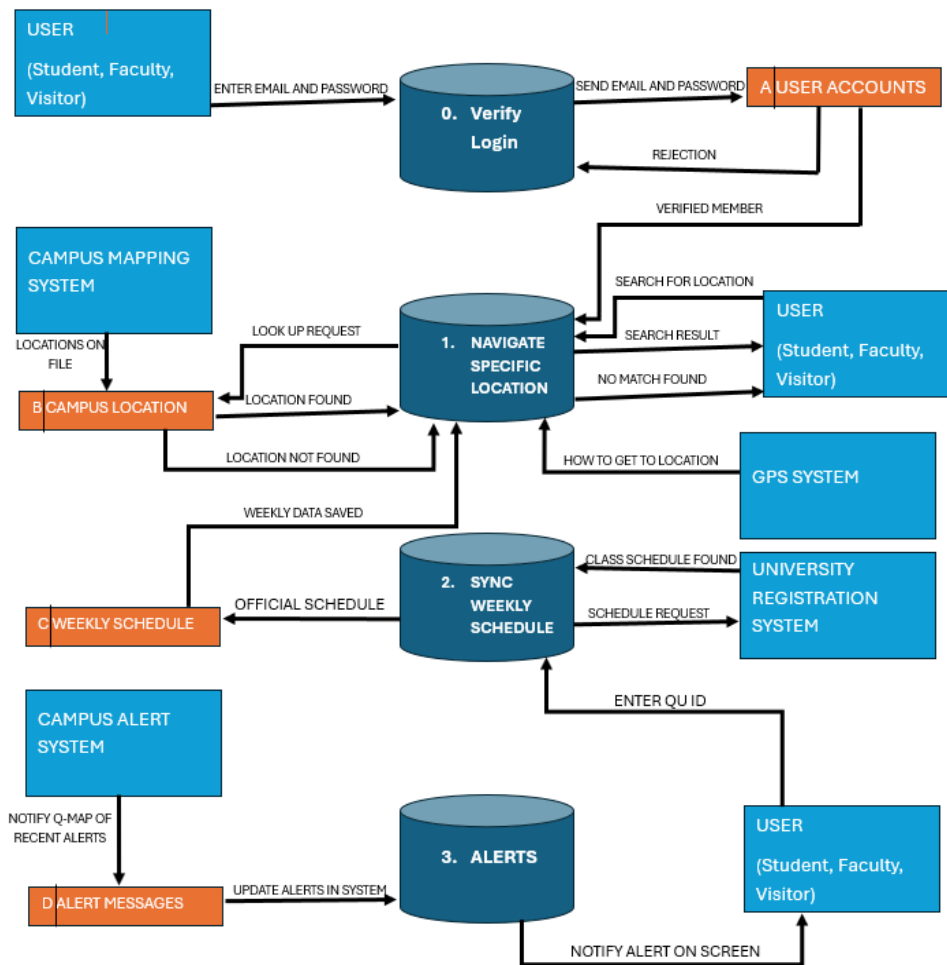
The location will only be viewable to a specific user, and no other user will be able to view any location other than their own. Any and all confidential student information, such as their student login credentials, will remain stored and protected from malicious activity. No user shall be allowed to access the system without being prompted to sign in first, using their QU login credentials, or by selecting the visitor option. The system will automatically log users out of the app should the user exit the application in any way.

Cultural and Political:

Our system must and will comply with Quinnipiac University's regulations and policies. Additionally, our system must and will comply with local, state, and federal law and regulations.

Process Models (DFD)

Q-MAP PROCESS MODELS (DFD)



Shown above is the data flow diagram for the Q-MAP application. This process model illustrates how data moves through the system and how each component interacts to support campus navigation, schedule syncing, and real-time alerts. The flow of this diagram centers around the four major processes that include, Verify Login, Navigate Specific Location, Sync Weekly Schedule, and Alerts. These processes convert user inputs into outputs by performing the necessary functions or activities within the system. Supporting these processes are the data stores, represented as long rectangles, which act as repositories where information is stored, retrieved, and updated. For Q-MAP, the primary data stores include User Accounts, Campus Location, Weekly Schedule, and Alert Messages. External entities, like the Campus Mapping System, University Registration System, GPS System, and Campus Alert System, provide additional data that flows into or out of each process. The final component of this model is the set of data flows, which represents the transfer of information between users, systems, and data stores. Together, these elements illustrate how Q-MAP processes login credentials,

navigational requests, schedule data, and campus alerts to deliver a functional and integrated experience for all users.

Process Description

The Q-MAP application is supported by four main processes: Verify Login, Navigate Specific Location, Sync Weekly Schedule, and Alerts. These processes work together to authenticate users, help them find campus buildings and rooms, keep academic schedules updated, and deliver real-time campus notifications. The first process, Verify Login, checks user credentials against the User Accounts data store to grant access to the system. Navigate Specific Location then allows users to search for buildings or rooms by referencing the Campus Location data store and the Campus Mapping System, while also using GPS data to generate directions if a location is found. The Sync Weekly Schedule process retrieves official class schedules from the University Registration System using the user's QU ID and stores the information in the Weekly Schedule data store for future use. Finally, the Alerts process updates and displays important campus notifications by receiving alert data from the Campus Alert System and storing it in the Alert Messages data store. Together, these processes allow Q-MAP to provide accurate navigation, schedule management, and timely alerts to the Quinnipiac community.

Process 0: Verify Login

DO RETRIEVE login input from the user, SEND credentials to the User Accounts data store, and VALIDATE them by checking for a match and active status. IF the credentials are verified, SET the session as logged in and RETURN a login confirmation. OTHERWISE, RETURN an error message and prompt the user to retry. UNTIL End-of-File.

Process 1: Navigate Specific Location

DO RETRIEVE the user's search request for a building, room, or point of interest, SEND a lookup request to the Campus Mapping System, and ACCESS the Campus Location data store for matching location details. IF a location is found, REQUEST route information from the GPS System, COMPUTE directions, and DISPLAY the map and navigation steps. OTHERWISE, RETURN a "No match found" message to the user. UNTIL End-of-File.

Process 2: Sync Weekly Schedule

DO RETRIEVE the user's sync request using their QU ID or portal credentials, SEND a schedule request to the University Registration System, and RECEIVE class schedule data. STORE the official schedule information in the Weekly Schedule data store and CONFIRM the sync success to the user while making the schedule available for navigation features. UNTIL End-of-File.

Process 3: Alerts

DO RECEIVE alert data from the Campus Alert System, UPDATE the Alert Messages data store with the new alert and its status, FILTER the alert based on user role, location, and relevance, and SEND and DISPLAY the notification on the user's device. UNTIL End-of-File.

Data Model

Data Store Report:

The data models presented below form the foundation of the Q-MAP data dictionary. This essential resource organizes the system’s data objects and provides a structured reference for how information is stored, accessed, and maintained within the application. The first table outlines each Q-MAP data store, Alert Messages, Campus Location, User Accounts, and Weekly Schedule, along with their corresponding data elements and designated primary or foreign keys. These repositories act as digital containers that manage the key information needed for campus navigation, user authentication, scheduling, and real-time safety notifications. The composition column lists the attributes that define each data store. Each attribute represents critical facts about the system’s data and supports the processes that rely on them. After examining each data store, we identified the essential elements required for the Q-MAP system to function effectively, as well as the relationships between the objects to ensure accurate data retrieval, consistency, and integrity across the application.

Data Store Name	Composition
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Alert Messages	Affected_Area Alert_DateTime Alert_ID (pk) Alert_Message Alert_Status Alert_Type
Campus Location	Building_Name Building_Type Floor_Number Last_Updated Latitude Location_ID (pk) Longitude Operating_Hours Room_Number
User Accounts	Account_Status Last_Login QU_ID (pk) User_Email User_FName User_LName User_Role
Weekly Schedule	Building_Name Course_Days Course_Name Course_Time Instructor_Name QU_ID (fk) Room_Number

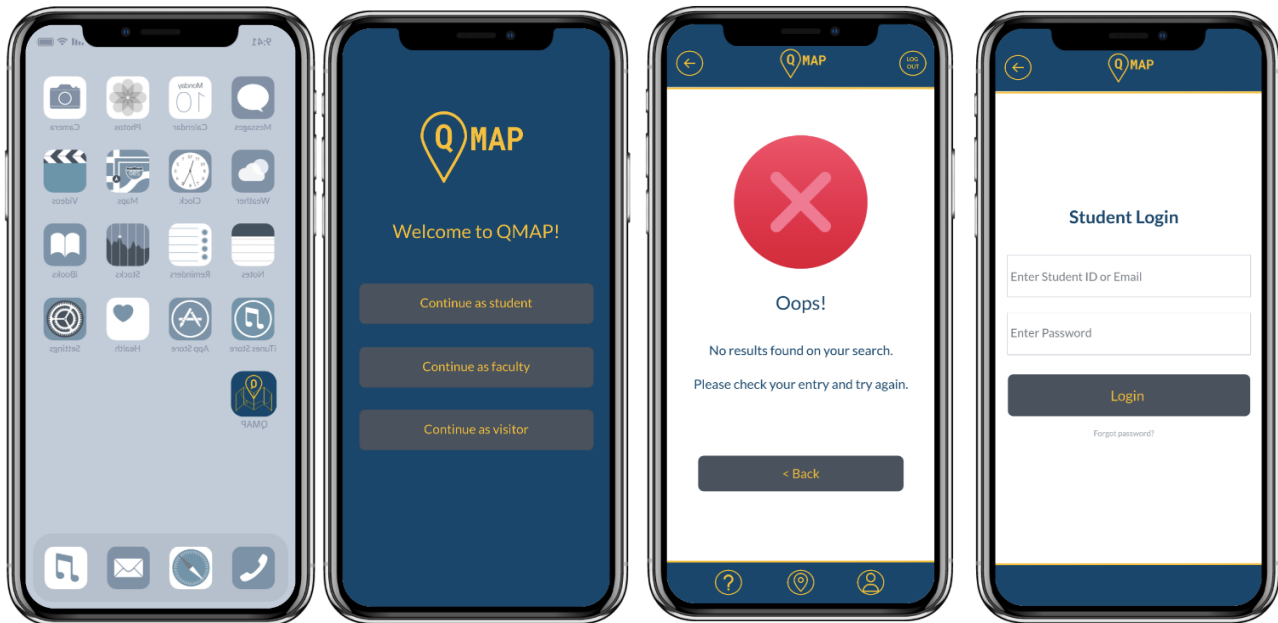
Data Dictionary

Data Element Report:

The second component of creating the Q-MAP data dictionary is the data element report, which builds upon the data store report by listing all attributes in alphabetical order along with their descriptions, data types, and defined lengths. This report outlines every data element that the system interacts with, including identifiers, location attributes, user information, scheduling details, and alert-related fields. Primary and foreign keys are clearly identified to indicate how records are uniquely established and connected across the system. By detailing elements such as building names, course information, alert data, user credentials, and GPS-based coordinates, the report highlights the full scope of information required for Q-MAP to function. This reference provides developers with a clear framework for understanding data attributes, their relationships, and their significance within the application. The data element report ensures consistency and accuracy throughout the development process and supports the system's ability to manage navigation, scheduling, user accounts, and real-time alerts effectively.

Data Element Name	Description	Type	Length
Account_Status	Current status of the user account (Active, Disabled, Pending)	Text	20
Affected_Area	Area or region impacted by the alert	Text	60
Alert_DateTime	When the alert was issued	Alphanumeric	25
Alert_ID (pk)	Unique identifier for each alert record	Number	10
Alert_Message	Content of the alert displayed to users	Text	100
Alert_Status	Current status of the alert (Active, Cleared)	Text	20
Alert_Type	Type of alert (Weather, Safety, Emergency)	Text	40
Building_Name	Full name of the building	Text	50
Building_Type	Category of building (Academic, Residential, Dining, etc.)	Text	40
Course_Days	Days of the week the class meets	Text	15
Course_Name	Title or name of the course	Text	60
Course_Time	Time interval when the class meets	Text	20
Floor_Number	Floor where the room is located	Number	2
Instructor_Name	Name of the course instructor	Text	50
Last_Login	Date/time the user last accessed the system	DateTime	—
Last_Updated	Date/time the building/location data was refreshed	DateTime	—
Latitude	GPS latitude coordinate	Decimal	6
Location_ID (pk)	Unique identifier for each location on campus	Number	10
Longitude	GPS longitude coordinate	Decimal	6
Operating_Hours	Hours during which the building or room is open	Text	50
QU_ID (fk/pk)	Unique identifier for an individual user	Number	7
Room_Number	Specific room identifier	Text	10
User_Email	User's registered email address	Text	60
User_FName	The first name of an individual user	Text	40
User_LName	The last name of an individual user	Text	40
User_Role	Indicates user type (Student, Faculty, Visitor)	Text	20

User Interfaces



When you first open the Q-MAP application, you are greeted with a clean and simple home screen that displays the app icon on the user's mobile device. Once the user taps the Q-MAP icon, the welcome screen appears with three options: "Continue as Student, Continue as Faculty, and Continue as Visitor". In our current prototype, only the student login workflow has been fully developed. Because the faculty and visitor pathways are still under construction, selecting either of those options triggers an error screen. This screen notifies the user that the requested feature is not yet available and prompts them to return to the previous page.

However, when a user selects the "Student" option, the system correctly advances to the student login page. Here, students are asked to enter either their Quinnipiac ID or email along with their password. After entering valid login information, they can continue into the app and access the rest of Q-MAP's features.

This login process ensures that students are authenticated before using the navigation, schedule syncing, and alert functionalities within the application. The error page for faculty and visitor access, combined with the functional student login process, clearly illustrates which portions of the prototype are currently operational and which ones are scheduled for future development.



When a student logs into Q-MAP successfully, the first screen they are brought to is the campus alert notification page. This alert appears immediately to ensure that users are aware of any safety concerns or important campus updates the moment they enter the application. The alert page is designed with a large icon and clear message so students can quickly understand the situation. For example, if there has been a coyote sighting or other safety-related incident near the user’s location, the app displays a warning and advises caution. Once the user reads the message, they can click the “OK” button at the bottom of the screen to proceed.

After acknowledging the alert, the user is taken to the main navigation page of the application. This serves as the central hub of Q-MAP, giving students access to campus maps, building details, and route guidance. The first view shows an interactive campus map with zoom controls and a location marker representing the user’s current position. At the bottom of the screen, the search bar allows students to enter the location they want to navigate to. Users can type a destination, select recent locations, or choose from preset options like “Home, Work, Dining, or Parking”. As they type, the suggestions update in real time, making it easy to find academic buildings, dining areas, or other campus facilities.

The navigation page also includes clear icons in the bottom bar for quick access to help, alerts, and the user’s profile. Overall, this flow ensures that students are informed about campus safety before seamlessly transitioning into the primary navigation features of the app.



When a user selects the help icon located on the bottom navigation bar, they are brought to the Q-MAP Help Center screen. This page is designed to support students by answering common questions and directing them toward additional assistance if needed. At the top of the screen, users see a highlighted message with contact information for the Q-MAP support crew, ensuring that students experiencing larger issues can quickly reach a team member by phone or email. Below this message is the Help Search Bar, where users can type specific questions regarding the app's features. To make the process even easier, recent searches are displayed underneath, offering quick access to frequently asked questions such as how to access the profile screen, how to view the weekly schedule, how to locate certain buildings, and how to troubleshoot navigation or error messages. This screen serves as a simple and intuitive support resource that helps users navigate Q-MAP independently.

When the user taps the profile icon on the bottom navigation bar, they are brought to their personalized profile and weekly schedule page. This section is directly tied to the student's QCard information, displaying their name, QU ID number, assigned room number, and their academic advisor. At the top of the screen, students have the option to edit their profile or switch between profiles if necessary. The lower portion of the screen shows the student's weekly routine, automatically populated from their university registration data. Each day of the week presents the student's class times, meeting locations, and room numbers, allowing them to quickly view and organize their daily schedule. This feature ensures that students have an easily accessible and organized overview of their academic commitments directly within the app.

References

“Q-Map.” *Marvel*, marvelapp.com/prototype/1cai58ej. Accessed 18 Nov. 2025.